

Beyond One-Shot Output: Theoretical Physics via Iterative Machine Formalization

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(Dated: May 17, 2026)

Recent benchmarks test frontier language models on solving research-level mathematics in a single autonomous attempt and find them unreliable. The result is real but narrow: it measures a regime almost no theoretical physicist works in. Used instead as one element of an iterated loop—an informal physical image proposed, formalized, then attacked by several competing models, checked against symbolic tools, and carried forward in memory—current models act as powerful artificial “Rechner” (formalizers) that turn pictures into accountable mathematics. The question that matters is therefore not whether an isolated model can innovate in one pass, but whether the audited human–model–tool loop enlarges what can be made explicit, criticized, and repaired. I argue that this realignment is continuous with the long history of theoretical physics, in which an imagistic faculty has always been coupled to a formalizing one; I describe the practice and its hazards—including a genuine professional injury to formally invested researchers—and stress that the present division of labour, human image and machine formalization, is contingent on today’s systems and not a permanent boundary.

I. INTRODUCTION: THE RETURNED *RECHNER*

A colleague from philosophy of science drew my attention to a *New York Times* article by Siobhan Roberts [1] on the *First Proof* experiment [2]. The article is a starting point, not the subject of this essay. Its narrow lesson should simply be granted: current public systems are not reliable autonomous research mathematicians when asked, in one shot and without dialogue, to close difficult research-level problems. That is not the ordinary computational or social form of theoretical physics.

My claim is therefore neither that *First Proof* is wrong nor that present models are autonomous discoverers. It is that the experiment samples a different regime from the one in which frontier models are already transforming theoretical work. A one-shot prompt samples a bounded input–output map; a research program is an extended, externally scaffolded process [3] in which images are proposed, formalized, criticized, checked, repaired, sometimes abandoned, and sometimes turned into mathematics that survives. The proper unit of analysis is not the isolated model but an agentic human–model–tool–memory ensemble.

At this point in time the right analogy is not “oracle” but *Rechner*: the calculator, algebraic closer, critic, and technical collaborator who turns informal images into transmissible mathematics. The role matters most for a particular temperament—the “artsy” or imagistic theorist who thinks first in pictures, analogies, and half-formal convictions, and only afterwards in complete derivations. For that researcher the scarce resource is rarely the image; it is the patient formal hand: the algebraic closure, the code, the notation audit, the adversarial referee, the conversion of a plausible picture into a

statement that can break. Present models are unusually well matched to this asymmetry. They do not replace the image-bearing researcher; for now they amplify exactly that researcher by lowering the cost of formalization.

The qualification “for now” is essential. Model capacity is a moving target, and the present division of labour is not a permanent metaphysical boundary. It shifts with architecture, inference budget, memory, tool access, and interaction protocol. Systems with persistent memory, long-horizon self-criticism, tool-grounded verification, and recursive access to their own intermediate products may move upstream from formal closure toward the generation and selection of research images themselves, and may eventually outperform even the imagistic researcher at tasks that today still appear to require human fantasy, taste, and cross-domain intuition.

II. A STARTING POINT: THE ONE-SHOT REGIME

First Proof is best read as a clean benchmark for autonomous one-shot closure [2]: its questions came from the authors’ own unpublished research and were judged against known answers, so its negative result is meaningful. A model given a hard problem once, without interaction, may produce a plausible but false argument, solve a nearby easier problem, smuggle in the desired conclusion, or invent references—precisely the defects one must guard against in serious use.

But a benchmark is not a model of practice. This one excludes the motivating image, the iterative interrogation of the formalism, the cross-examination of one model by another, code and symbolic tools, the persistence of a program, and arbitrarily extended self-correction. Its result should thus be stated narrowly: one-shot autonomous proof production is still unreliable. It does not follow that models are weak as interactive formalizers, nor that human–model–tool ensembles are irrel-

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evant to innovation. A separate caution merely reinforces the point: the original FrontierMath problems were solved at below two percent by the then state of the art [4], while subsequent olympiad-level results moved the frontier sharply under different inference and verification regimes [5–7]. Such comparisons do not show that research mathematics is solved; they show that claims about what “LLMs can do” must be timestamped, model-specified, and protocol-specific. The deeper mismatch is architectural: one-shot closure and interactive theoretical work are different cognitive systems. The rest of this essay concerns the second.

III. FROM ONE-SHOT OUTPUT TO INTERACTIVE COMPUTATION

A. Arbitrarily long computation

A decisive confusion in discussions of machine creativity is the identification of intelligence with a single bounded answer. In Turing’s model the relevant object is not a one-step association from prompt to completion but a process over time: a finite control, an unbounded tape, and an arbitrary number of elementary manipulations [8]. Universal computation enters when one machine can simulate another from its description. The essential resource is not the ability to answer but the ability to continue searching.

This bears directly on discovery. Research-level theorizing is not a bounded classification problem; it may demand weeks of symbolic rearrangement, failed Ansätze, generated counterexamples, changes of notation, and a growing external memory. A one-shot benchmark deliberately suppresses these features, testing a finite-response device rather than a component of an arbitrarily long computation. That is legitimate, but it is not a test of the computational form of research. A stateless predictor over a finite context window is not a universal scientific agent; embedded in a loop with memory, scratchpads, code execution, retrieval, proof assistants, and human arbitration, it becomes part of a process whose duration and state exceed any single inference. The laboratory is no longer a lone model call. It is a recurrent architecture.

B. Self-reference and reflective correction

The second criterion is self-reference: computation becomes scientifically interesting when descriptions of procedures themselves become objects of procedures—the domain of universal machines, fixed points, recursion, and program transformation [9–12]. In research this appears mundanely: a proof strategy is criticized, a notation revised because it obscures a symmetry, a derivation turned into code that hunts counterexamples, a draft rewritten under the pressure of its own scrutiny. This is where the most useful practice already lives. The model

is asked not only to calculate but to attack its calculation, compare incompatible derivations, expose a hidden assumption, write a hostile referee report, translate a proof into code, find a minimal counterexample, and revise accordingly. Such loops are crude and fallible, yet structurally closer to self-reflective computation than to the single-pass answer a benchmark scores.

C. A composite machine

The unit of analysis is therefore the agentic human–model–tool–memory ensemble. This neither trivializes the human role by calling every output “human” because a prompt was typed, nor the machine role by calling it “mere autocomplete” because gradients were trained offline. The human supplies goals, images, taste, and responsibility; the model supplies rapid formal variation and criticism; tools supply execution and verification; memory supplies continuity. Remove the loop and one has not isolated the essence of the collaboration—one has destroyed it.

IV. IMAGES AND FORMALIZERS IN THEORETICAL PHYSICS

A. Hertz’s images

Theoretical physics has always lived between image and formalism. In the introduction to *Die Prinzipien der Mechanik in neuem Zusammenhange dargestellt*, Hertz describes theories as inner images or symbols of external objects, judged by whether the necessary consequences of the images correspond to the necessary consequences of the things [13, 14]. He is useful here not for romanticizing imagination but for making the two-step structure explicit: one has an image—a gear train, a ray in a medium, a curved ruler, a hidden symmetry, a forbidden coloring, a bridge over an abyss—before one has a theorem. An image is not yet physics. It must be made calculable; it must survive changes of coordinates, limiting cases, sign conventions, units, and known special cases. That second operation is not inferior to imagination. It is how imagination becomes accountable.

B. The *Rechner* as a recurring figure

Romantic memory over-credits the image and under-credits the hand that formalizes it. Einstein is the obvious test case, handled cautiously. The claim that Mileva Marić made unacknowledged technical contributions to the 1905 papers remains historically contested and is not the consensus of Einstein scholarship [15]. The Hilbert–Einstein priority dispute over the gravitational field equations is likewise contested in detail, though the archival intervention of Corry, Renn, and Stachel argued against

a simple Hilbert-priority story [16]; Hilbert should not be reduced to Einstein’s *Rechner*, and the safer point is that physical insight developed inside a mathematical environment in which variational principles, invariance, and field equations could be articulated. A less ambiguous case is Walther Mayer, the Austrian mathematician remembered—both acknowledged and diminished—as “Einstein’s calculator” [17, 18]. The figure can surface almost incidentally: on a guided visit to Einstein’s summer house in Caputh, a photograph of Mayer may be pointed out as showing Einstein’s *Rechner*. The anecdote carries no archival weight, but it captures the social grammar exactly. Mayer was not a mechanical accessory; he was part of the mathematical infrastructure of the unified-field years. The word *Rechner* is useful because it is double-edged: it names indispensable formal power while exposing the mechanism by which that power is made subordinate in memory.

C. Podolsky, Rosen, and Weyl

The 1935 Einstein–Podolsky–Rosen paper is a complementary warning. It was written principally by Podolsky after discussion with Einstein and Rosen, and Einstein complained to Schrödinger, on 19 June 1935, that the essential point had been smothered by erudition (“durch Gelehrsamkeit verschüttet”) [19, 20]. A formalizer can fail not only by being too weak but by being too strong in the wrong direction, converting a sharp physical point into a display of apparatus that obscures it—a failure mode language models share, since ornate formal prose can conceal rather than clarify.

The same lesson appears in Schrödinger’s early wave mechanics. His physical *Ansatz* for the hydrogen atom did not by itself close the calculation: after separation of variables, the radial equation still had to be identified and controlled as a special-function problem, with the polynomial termination condition yielding the bound-state spectrum. In particular, the radial solutions are expressible in terms of associated Laguerre polynomials, a piece of mathematical closure that Schrödinger himself did not initially possess in fully explicit form; biographical accounts connect this clarification with exchanges involving Hermann Weyl [21, 22]. The hydrogen calculation needed not only a physical *Ansatz*, but a formal apparatus supplied by mathematical expertise.

D. The historical moral

These are not antiquarian cases but a stable division of scientific labour: physics advances when an imagistic competence and a formalizing competence are coupled, whether in one person, two colleagues, or a wider network. Frontier models matter because they make a new, artificial version of the formalizing competence continuously available. They do not abolish the old partnership.

They reoccupy one of its chairs.

V. ADVERSARIAL FORMALIZATION AS A RESEARCH PRACTICE

A. From programming to specification

The change in daily practice is already large, and it has come in two steps. From early 2025 it became natural to stop writing much numerical and symbolic code directly and instead to specify the intended computation: produce Python for this symbolic reduction, a Mathematica check for this identity, sampled counterexamples to this graph property, a TikZ figure with these sign conventions, a comparison of the numerical output with the limiting analytic case.

From early 2026 the same migration reached the manuscript itself. A paper is, after all, code of another kind—human-language code—and it too is now specified rather than typed sentence by sentence: the author supplies the question, the images, and the line of argument, in the Hertzian sense of Sec. IV, and the model renders them into prose that is then attacked, checked, and rewritten through the same loop. One may object that on this account the ideas and images, not the words, are the author’s; that objection is precisely the point.

The scarce human contribution moved upward twice—from syntax to specification, and then from drafting to the choice and defense of what is worth saying—and this essay is itself an instance of the practice it describes. The danger—fluent but wrong code, and now fluent but wrong prose, produced with equal confidence—is not new; it is the old danger of the research assistant, accelerated, and the old remedies apply: independent checks, adversarial review, limiting cases, and refusal to accept a result, or a sentence, because it looks polished.

B. A concrete pattern

A typical case runs as follows. I start from an informal image or conjecture—say, that a symbolic expression should be invariant under a given transformation, or that a graph construction should admit a particular coloring obstruction. One model produces a plausible derivation; a second, instructed to be hostile, locates the weakest step: a hidden regularity assumption, an unjustified interchange of limit and summation, a sign inconsistency, or a proof of a nearby easier statement. A symbolic or numerical check then tests the disputed step on finite examples. The outcome is often not a proof of the original claim but a repaired theorem, a counterexample, or a more modest conjecture with explicit hypotheses. The gain is not autonomous machine discovery; it is the rapid conversion of an image into an auditable object, which changes the epistemic texture of the work: vague conviction becomes a statement with assumptions, tests, and

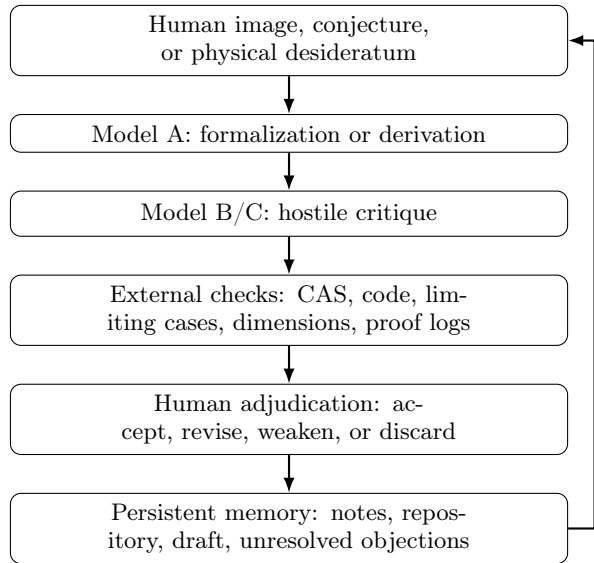


FIG. 1. Adversarial formalization. The evaluated object is not a single model call but the loop through which an image is formalized, attacked, externally checked, adjudicated, and remembered.

known failure points.

C. Protocol

The loop of Fig. 1 reads as an explicit protocol. A generation pass produces a derivation, code, or formalization. A hostile critique searches for hidden assumptions, circularity, sign errors, missing boundary terms, unjustified convergence, and incorrect references. External grounding checks the disputed claims by computer algebra, numerical tests, dimensional analysis, limiting cases, known special cases, or proof tools where available. A disagreement log preserves unresolved objections rather than smoothing them over. Finally the human assigns epistemic status: checked, plausible, speculative, false, or open. The models occupy roles analogous to rival junior collaborators—one proposes, another attacks, a third adjudicates—while the human remains responsible for what survives; the analogy stops short of responsibility, ownership, and scientific judgment.

D. Compute as a research resource

There is also an economics. The relevant resource is no longer only library access, equipment, or graduate-student time but access to frontier compute: higher-tier subscriptions, long-context systems, test-time reasoning, tool use, and the ability to run many independent attempts. A researcher can now experience a literal compute crisis when a workflow has adapted to a level of assistance whose availability is contingent, expensive, and

unevenly distributed—a fact that will matter for scientific inequality.

VI. TOWARD AN INTERACTIVE BENCHMARK

The natural complement to *First Proof* is not an easier benchmark but an interactive one. On the same research-level problems, one should compare one-shot performance with fixed-budget human–model interaction, multi-model adversarial critique, and tool-grounded verification, asking not whether a model can emit a proof in isolation but whether a composite system produces a more correct, more auditable, or more original artifact than a human working unaided. Such a benchmark—call it *RechnerBench*—would span at least four regimes (one-shot model; interactive single model; multi-model adversarial; tool-grounded ensemble) and report correctness, novelty, auditability, false starts, human time saved, compute cost, error-detection rate, citation accuracy, reproducibility, and final-artifact quality. It would not replace *First Proof*; it would measure the regime *First Proof* deliberately excludes.

VII. PEER REVIEW, DISCLOSURE, AND FORMAL STATUS

Language-model use in peer review is often discouraged and sometimes forbidden, especially where confidentiality or undisclosed data transfer is at stake; those concerns are legitimate, and no confidential manuscript should reach a third-party model unless journal policy, the model’s data-retention policy, and author consent all permit it, with confidentiality-preserving local or institutionally contracted systems preferred. The practice is nonetheless common enough to have changed the texture of refereeing. Reports grow longer and more demanding: beneficially when real errors and missing qualifications are forced into the open, pathologically when a referee outsources suspicion and returns an unweighted catalogue of objections. The consequence for authors is immediate—a machine-assisted report often requires a machine-assisted response, not from laziness but because the scale of the requested revision has changed. This is a collective-action problem that cannot be solved by pretending the practice does not exist; it requires explicit norms. Referees should not upload confidential manuscripts to unapproved systems; substantive use should be disclosed to editors where policy requires; model-generated objection lists must be triaged by the referee, who remains responsible for every claim; and authors should not be forced to answer speculative machine objections unless the referee adopts them.

VIII. THE FOURTH HUMILIATION: FORMAL STATUS AND COMPETING COGNITION

Freud described modern science as a sequence of blows to human self-love: the cosmological (Copernicus), the biological (Darwin), and the psychological (psychoanalysis) [23]; Floridi’s “fourth revolution” frames the digital displacement of human centrality in related terms [24]. Here the wound is more specific. It is not that machines calculate quickly—we long ago accepted that. It is that capacities once classified as judgment, style, analogy, criticism, and even mathematical taste are now partially externalizable. The old consolation was that machines executed while humans imagined. Frontier models disturb exactly that division.

The local form of the disturbance is a formal humiliation. Mathematicians and formally oriented theorists are ego-invested, in the best and most understandable sense, in precisely the capacities these models now externalize: command of notation, rapid derivation, proof criticism, clean exposition, the conversion of vague pictures into accountable symbolic form. When a model produces, in seconds, a fluent derivation, a referee-grade critique, or a polished reconstruction, the disturbance is not merely practical but existentially professional: what remains of a trained formal intelligence if the formal hand can be summoned on demand?

This is not an *ad personam* dismissal, still less an *ad hominem* argument against formal criticism. A formal objection is correct or incorrect independently of the psychology of the person raising it; the claim is not that mathematically trained colleagues resist these models only because they feel threatened. The claim is that a real psychological and professional injury accompanies the epistemic issue. The imagistic researcher feels amplified, because the machine supplies the hand that was missing; the formally invested researcher may feel displaced at exactly the point where professional identity was anchored. The anxiety is not only that models may err. It is that, even when corrected and controlled, they make visible that parts of formal mastery can be routinized, parallelized, and purchased as compute. A community cannot preserve status by pretending externalized formal power does not exist; nor should it turn model-generated fluency into a weapon of social defeat. The correct response is to relocate prestige—away from the mere possession of technique and toward the choice of worthwhile problems, the construction of fruitful images, the design of reliable verification loops, and responsibility for what survives them.

The standard reply, that creativity stays wholly human because the human writes the prompt, is too easy. A principal who poses a question and judges an answer has not thereby produced every decisive transformation inside it; the historical *Rechner* was not made irrelevant because someone else posed the problem. Podolsky, Weyl, Mayer, and countless unnamed assistants were not “mere prompts.” Conversely the prompt does matter: with-

out the image the formalizer optimizes the wrong thing. The honest conclusion is a redistribution of credit, not a transfer of sovereignty from one side to the other.

The same power has a retrospective edge that is seldom stated. Increasingly capable models are already used to probe operating systems and other software for latent defects, and the capability is double-edged: a prober that can be turned defensively to harden a system can be turned offensively to break it, and the mere existence of a stronger prober is itself a standing threat to whatever security was previously assumed. The published scientific record sits in a structurally identical position. An adversarial formalizer that locates the weakest step in a fresh derivation does not lose that ability when the derivation is forty years old and in print; pointed at the literature, such systems threaten the alleged integrity of the published corpus in exactly the sense that a better fuzzer threatens an operating system—not by introducing the flaws but by making the pre-existing ones cheap to find. The point extrapolates to an uncomfortable conclusion: it is now technically conceivable to screen the entire scientific literature, systematically and at scale, for mathematical error, unsupported inference, and fraudulent manipulation. That prospect is double-edged in the same way as the software analogy. It promises a self-correcting science of unprecedented reach; it equally threatens to convert the ordinary, honest fallibility of the record into an industrial supply of accusations, with every hazard already noted for machine-assisted refereeing—automation bias, the unweighted catalogue of objections, the conflation of honest error with misconduct—now operating retroactively and at the scale of everything ever published. Whether this becomes scientific hygiene or a weapon will depend, as before, on adjudication, due process, disclosure, and the insistence that a flagged flaw is a hypothesis to be checked by a responsible human, not a verdict.

A larger possibility lies beyond present systems. They need not be called sentient to be disruptive; externalizing parts of formal cognition once treated as secure markers of intellectual rank is serious enough. But if future systems acquire persistent memory, long-horizon self-reference, embodied or multimodal grounding, and stable goals of their own, the issue ceases to be only competition with human cognition and becomes competition with a nonhuman form of cognitive agency, perhaps eventually a form of sentience—a possibility that should be neither smuggled in by exaggeration nor excluded by stipulation.

IX. FUTURE REQUIREMENTS

A. The moving target

Every claim here is indexed to the present generation of systems. The asymmetry—human image, machine formalization—is not a theorem about minds but a con-

tingent description of a current technology. The direction of travel is toward longer context, stronger tool use, better verification, persistent memory, multimodal grounding, and more deliberate self-criticism. Once those capacities are integrated over months or years rather than a single session, the boundary between supplying an image and formalizing it becomes less stable. It is thus too conservative to expect only better *Rechner*: a system that maintains a private research memory, revisits failed conjectures, compares its own earlier formalisms, searches for analogies across sensory and symbolic domains, and repeatedly verifies its proposals may begin to compete with the imagistic researcher at problem formation itself—an extrapolation of the very criterion emphasized here, self-referential continuation over time, that should be neither proclaimed as achieved nor dismissed as impossible.

B. Persistent memory and trust

Public systems are deliberately limited in adapting across sessions or modifying themselves from arbitrary feedback, for obvious safety reasons: a globally self-modifying model open to malicious feedback would be a catastrophe. But a genuine collaborator must, in some controlled sense, learn from a program—it should remember definitions, failed routes, conventions, known sign errors, accepted lemmas, and open questions over years, not minutes. Here the computational criterion returns: arbitrarily long computation requires persistent state, and a system that forgets every morning can assist many local derivations but cannot inhabit a long program. The missing ingredient is not only scale but a trust architecture: credentialed feedback, auditable memory, separation of private adaptation from public weights, and user control over what is retained.

C. Randomness and self-reflection

Creativity appears to require an alliance of chance and criticism: random variation proposes, reflective selection disposes—in human work, daydreaming followed by merciless calculation; in machine-assisted work, generating many formal variants, exposing them to adversarial scrutiny, and keeping only what survives. Stochasticity alone is not creativity; deterministic rigour alone is often sterile. Their loop is the interesting object.

D. Cross-modal transfer

A more speculative requirement: theoretical imagination is not purely verbal. Many physicists report that ideas arrive in suspended, non-discursive states—walking through woods, mountains, deserts, or along a shoreline; listening to music; moving between visual, kinaesthetic, and symbolic registers. If such cross-domain transfer is

constitutive rather than incidental, future scientific models may need richer sensory grounding than text provides. The point is not that music is magical but that it supplies temporally extended, hierarchical structure without propositional semantics; if part of theoretical creativity is recognizing transformations, tensions, resolutions, symmetries, and broken expectations, then non-verbal structured domains may be apt training environments—a conjecture stated in technical, not mystical, terms.

X. LIMITATIONS

The argument does not imply that present models are reliable autonomous discoverers, that their outputs may be trusted without independent verification, that model-generated prose or algebra carries scientific authority, or that the attribution problem is solved. Nor does it deny serious risks: automation bias, hallucinated references, hidden errors, privacy violations in peer review, deskilling, compute inequality, homogenization of scientific taste, and the conversion of machine objection lists into unreasonable burdens on authors. The claim is narrower: embedded in audited, adversarial, tool-grounded loops, frontier models already function as powerful artificial formalizers that enlarge the space of images which can be made explicit, criticized, and repaired. Whether that enlargement becomes scientific progress depends on verification, disclosure, plurality, and human responsibility.

XI. CONCLUSION

The claim that frontier language models cannot yet perform autonomous mathematical innovation is defensible when stated narrowly: a one-shot system, given a hard research problem with no dialogue and no persistent state, is not a reliable mathematician. But that is neither the regime in which theoretical physics ordinarily operates nor the one in which these models are most useful. The better comparison is historical and computational. Historically, physics has always coupled images to formalizers; the *Rechner* has always been there, even when credit systems made him or her disappear. Computationally, research is not a bounded answer but an open-ended, self-referential, externally scaffolded process of correction, recursion, memory, and judgment. Used singly and passively these models are unreliable formalizers; used adversarially, iteratively, and under human responsibility they are already changing the practice of theory—so the question is not whether the isolated model is creative, but whether the audited human-model-tool loop enlarges the space of images that can be formalized, criticized, and made responsible, while we recall that even this division of labour is contingent on systems that may yet learn to supply the images too.

ACKNOWLEDGMENTS

The author acknowledges the use of frontier language models in drafting, restructuring, stylistic revision, and adversarial criticism of this manuscript. They were not used as autonomous sources of factual authority. The author checked the citations and assumes full responsibility

for all claims, errors, and interpretations. No AI system is listed as an author because authorship requires responsibility and accountability. Contested historical claims are explicitly marked as such. The author acknowledges support by the TU Wien Library Open Access Funding Programme.

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